Parametric Equations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Eliminating the parameter**

Consider the curve $x = 2t + 1$, $y = t^2$.

a Eliminate the parameter to write y as a function of x. $t = \frac{x-1}{2}$, so $y = \frac{(x-1)^2}{4}$.

b Identify the curve. A parabola with vertex $(1, 0)$ opening upward.

(a) $t = \frac{x-1}{2}$, so $y = \frac{(x-1)^2}{4}$.; (b) A parabola with vertex $(1, 0)$ opening upward.

2 **[Foundational]** • **Target: Eliminating the parameter**

Eliminate the parameter for $x = 3\cos t$, $y = 3\sin t$, $0 \le t < 2\pi$, and describe the curve.

$x^2 + y^2 = 9\cos^2 t + 9\sin^2 t = 9$: a circle of radius 3 centered at the origin, traced counterclockwise.

3 **[Foundational]** • **Target: Eliminating the parameter**

For the line $x = t - 2$, $y = 2t + 3$:

a Find the points corresponding to $t = 0$ and $t = 2$. $t = 0$: $(-2, 3)$; $t = 2$: $(0, 7)$.

b Eliminate the parameter to write the Cartesian equation. $t = x + 2$, so $y = 2x + 7$.

(a) $t = 0$: $(-2, 3)$; $t = 2$: $(0, 7)$.; (b) $t = x + 2$, so $y = 2x + 7$.

4 **[Foundational]** • **Target: Projectile motion**

A projectile follows $x = 40t$, $y = -5t^2 + 30t$ (metres, seconds).

a When does the projectile land ($y = 0$, $t > 0$)? $-5t(t - 6) = 0$, so $t = 6$ s.

b Find the horizontal range. $x(6) = 240$ m.

c Find the maximum height. Peak at $t = 3$: $y(3) = -45 + 90 = 45$ m.

(a) $-5t(t - 6) = 0$, so $t = 6$ s.; (b) $x(6) = 240$ m.; (c) Peak at $t = 3$: $y(3) = -45 + 90 = 45$ m.

5 **[Foundational]** • **Target: Projectile motion**

A second projectile is launched with $x = 25t$, $y = -5t^2 + 40t$. Find its time of flight and maximum height.

Land: $-5t(t - 8) = 0$, so $t = 8$ s. Peak at $t = 4$: $y(4) = -80 + 160 = 80$ m.

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6 [Foundational] • Target: Ellipses and other parametric curves

Eliminate the parameter for $x = 4\cos t$, $y = 2\sin t$, and identify the resulting curve.

$\frac{x}{4} = \cos t$, $\frac{y}{2} = \sin t$: $\frac{x^2}{16} + \frac{y^2}{4} = 1$ — an ellipse with semi-axes 4 and 2.

7 [Foundational] • Target: Ellipses and other parametric curves

Eliminate the parameter for $x = t^2 - 1$, $y = t^3$ by expressing t in terms of x where possible, or by finding a relation between x and y directly.

From $x = t^2 - 1$: $t^2 = x + 1$. Squaring $y = t^3$: $y^2 = t^6 = (t^2)^3 = (x + 1)^3$. Relation: $y^2 = (x + 1)^3$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Finding the parametric derivative

For the curve $x = t^2$, $y = 4t$, find $\frac{dy}{dx}$ in terms of t using $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$.

$\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 4$: $\frac{dy}{dx} = \frac{4}{2t} = \frac{2}{t}$ (for $t \neq 0$).

9 [Intermediate] • Target: Finding the parametric derivative

For the curve $x = \cos t$, $y = \sin t$, find the slope of the tangent line at $t = \frac{\pi}{4}$.

$\frac{dx}{dt} = -\sin t$, $\frac{dy}{dt} = \cos t$: $\frac{dy}{dx} = -\cot t$. At $t = \frac{\pi}{4}$: slope $= -\cot \frac{\pi}{4} = -1$.

10 [Intermediate] • Target: More parametric practice

A Ferris wheel of radius 10 m has its center 12 m above the ground and completes one revolution every 40 seconds. Write parametric equations for the height y of a rider above the ground as a function of time t (seconds), assuming the rider starts at the bottom of the wheel.

Angular speed $\omega = \frac{2\pi}{40} = \frac{\pi}{20}$. Starting at the bottom means the vertical position begins at a minimum, matching $-\cos$: $y(t) = 12 - 10\cos\left(\frac{\pi}{20}t\right)$.

11 [Intermediate] • Target: More parametric practice

Two particles travel along paths $x_1 = t,\ y_1 = t^2$ and $x_2 = 4 - t,\ y_2 = 3t - 2$, sharing the same parameter t (time). Find the value of t at which they collide, and the point of collision.

Need $x_1 = x_2$: $t = 4 - t$, so $t = 2$. Check $y_1 = y_2$ at $t = 2$: $y_1 = 2^2 = 4$; $y_2 = 3(2) - 2 = 4$. Both match, confirming a genuine collision at $t = 2$, at the point $(2, 4)$.

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12 [Intermediate] • Target: Graphing a parametric curve after elimination

For $x = t$, $y = t^2 - 2$ (with t ranging over all reals), the eliminated Cartesian form $y = x^2 - 2$ is shown. Use the graph to describe the direction the curve is traced as t increases.

As t increases from $-\infty$ to ∞, the curve is traced from the upper-left, down through the vertex $(0, -2)$, then up to the upper-right — left to right along the parabola.

13 [Intermediate] • Target: Restricting the parameter domain

The curve $x = \cos t$, $y = \sin t$ for $t \in [0, \pi]$ traces only part of the unit circle. Describe which part, and state the domain and range restrictions this implies on x and y.

For $t \in [0, \pi]$, $y = \sin t \geq 0$, so only the UPPER half of the unit circle is traced (from $(1, 0)$ counterclockwise to $(-1, 0)$). This restricts $y \geq 0$ while x still ranges over $[-1, 1]$.

14 [Intermediate] • Target: Parametrizing a line segment

Write parametric equations for the line segment from $A(2, -1)$ to $B(6, 5)$, using $t \in [0, 1]$ (so $t = 0$ gives A and $t = 1$ gives B).

$x(t) = 2 + 4t$, $y(t) = -1 + 6t$ for $t \in [0, 1]$ (using $x = x_1 + t(x_2 - x_1)$, $y = y_1 + t(y_2 - y_1)$).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Parametrizing a line segment • Cross-Lesson: 1.03 Average Rate of Change

Using the parametrization above, find the point at $t = 0.25$ (one-quarter of the way from A to B).

$x(0.25) = 2 + 1 = 3$; $y(0.25) = -1 + 1.5 = 0.5$. Point: $(3, 0.5)$.

16 [Challenging] • Target: Parametric speed and motion review • Cross-Lesson: 2.01 Right-Triangle Trigonometry

A particle's position is $x(t) = 3t$, $y(t) = 4t$ for $t \geq 0$ (measured in meters, seconds). Find the particle's speed, and confirm it matches $\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$.

$\frac{dx}{dt} = 3$, $\frac{dy}{dt} = 4$. Speed $= \sqrt{9 + 16} = 5$ m/s (constant, matching straight-line motion at constant velocity).

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17 [Challenging] • Target: Finding the parametric derivative • Cross-Lesson: 2.02 Radian Measure and Arc Length

A population of bacteria, initially P_0 cells, grows according to the logistic model $P(t) = \frac{L}{1 + Ae^{-kt}}$, where t is in hours, L is the carrying capacity, and A and k are positive constants. At $t = 0$, the population is $P(0) = \frac{L}{3}$. After 2 hours, the population has doubled to $P(2) = \frac{2L}{3}$. Find the exact value of e^{2k} in terms of L and P_0.

The logistic model is given by $P(t) = \frac{L}{1 + Ae^{-kt}}$. At $t = 0$, $P(0) = \frac{L}{3}$. Substituting into the model: $\frac{L}{3} = \frac{L}{1 + Ae^{-k \cdot 0}}$ $\frac{L}{3} = \frac{L}{1 + A}$ $1 + A = 3$ $A = 2$. So the model becomes $P(t) = \frac{L}{1 + 2e^{-kt}}$. We are also given that $P(2) = \frac{2L}{3}$. Substituting into the model: $\frac{2L}{3} = \frac{L}{1 + 2e^{-2k}}$ $\frac{2}{3} = \frac{1}{1 + 2e^{-2k}}$ $2(1 + 2e^{-2k}) = 3$ $2 + 4e^{-2k} = 3$ $4e^{-2k} = 1$ $e^{-2k} = \frac{1}{4}$. We need.

18 [Challenging] • Target: More parametric practice • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

A radioactive isotope has a decay constant $k = 0.035\ \text{year}^{-1}$. (a) Find its half-life to the nearest tenth of a year. (b) If an initial sample contains $250\ \text{mg}$, how much remains after $40\ \text{years}$?

(a) Half-life calculation: $t_{1/2} = \frac{\ln 2}{k} = \frac{0.693147}{0.035} \approx 19.8\ \text{years}$. (b) Remaining mass at $t = 40$: $A(t) = A_0 e^{-kt} = 250e^{-0.035(40)} = 250e^{-1.4} \approx 250(0.246597) \approx 61.65\ \text{mg}$.

19 [Challenging] • Target: Graphing a parametric curve after elimination • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

A geometric sequence has its first term $a_1 = 5$ and its common ratio $r = 2$. If the sum of the first n terms of this sequence is $S_n = 315$, find the value of n.

The sum of the first n terms of a geometric sequence is given by the formula $S_n = a_1 \frac{r^n - 1}{r - 1}$, where a_1 is the first term and r is the common ratio. Given $a_1 = 5$ and $r = 2$, and $S_n = 315$, we can substitute these values into the formula: $315 = 5\frac{2^n - 1}{2 - 1}$ $315 = 5\frac{2^n - 1}{1}$ $315 = 5(2^n - 1)$ Divide both sides by 5: $\frac{315}{5} = 2^n - 1$ $63 = 2^n - 1$ Add 1 to both sides: $63 + 1 = 2^n$ $64 = 2^n$ Since $2^6 = 64$, we have $n = 6$. Let's check the choices.

20 [Challenging] • Target: Restricting the parameter domain • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Consider the complex number $z = r(\cos\theta + i\sin\theta)$. If $z^3 = -8$ and $r > 0$, and the principal argument of z is $\frac{\pi}{3}$, find the value of r.

We are given the complex number $z = r(\cos\theta + i\sin\theta)$, where $r > 0$. We are also given that $z^3 = -8$ and the principal argument of z is $\theta = \frac{\pi}{3}$. Using De Moivre's Theorem, we have $z^3 = r^3(\cos(3\theta) + i\sin(3\theta))$. We are given $z^3 = -8$, which can be written in polar form as $8(\cos(\pi + 2k\pi) + i\sin(\pi + 2k\pi))$ for any integer k. Equating the magnitudes, we get $r^3 = 8$. Since $r > 0$, we have $r = \sqrt[3]{8} = 2$. Equating the arguments, we get.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Plotting and Direction of Motion

For the parametric curve $x = 2t + 1, y = t^2 - 4$ for $-2 \le t \le 2$, find the initial point and terminal point.

a. **Initial:** $(-3, 0)$**, Terminal:** $(5, 0)$ ✓

b. Initial: $(1, -4)$, Terminal: $(5, 0)$

c. Initial: $(-3, 0)$, Terminal: $(1, -4)$

d. Initial: $(0, -3)$, Terminal: $(0, 5)$

At $t = -2$: $x = 2(-2) + 1 = -3$, $y = (-2)^2 - 4 = 0$. At $t = 2$: $x = 2(2) + 1 = 5$, $y = 2^2 - 4 = 0$.

22 [Foundational] • Target: Plotting and Direction of Motion

How is orientation (direction of motion) shown on a parametric graph?

a. **Arrows placed along the curve indicating the direction as the parameter t increases** ✓

b. By the slope of the curve

c. By the color of the line

d. By the quadrant of the initial point

Orientation arrows indicate the forward direction of motion as time/parameter t advances.

23 [Foundational] • Target: Plotting and Direction of Motion

Determine the direction of motion for $x = 3 - t, y = 2t$ as t increases.

a. **Leftward and upward** ✓

b. Rightward and upward

c. Leftward and downward

d. Rightward and downward

As t increases, $x = 3 - t$ decreases (leftward) and $y = 2t$ increases (upward).

24 [Foundational] • Target: Eliminating the Parameter (Algebraic)

Eliminate the parameter t from $x = t - 3, y = 2t + 1$.

a. $y = 2x + 7$ ✓

b. $y = 2x - 5$

c. $y = 2x - 7$

d. $y = x + 4$

Solve for t: $t = x + 3$. Substitute into y: $y = 2(x + 3) + 1 = 2x + 6 + 1 = 2x + 7$.

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25 [Foundational] • Target: Eliminating the Parameter (Algebraic)

Eliminate the parameter from $x = t^2, y = 3t$ for $t \geq 0$.

a. $x = \frac{y^2}{9}$ for $y \geq 0$

b. $y = 3\sqrt{x}$

c. **Both of the above are correct ✓**

d. $x = 9y^2$

$t = y/3 \Rightarrow x = (y/3)^2 = y^2/9$, or equivalently $y = 3\sqrt{x}$ for $x \geq 0, y \geq 0$.

26 [Foundational] • Target: Eliminating the Parameter (Algebraic)

Eliminate the parameter from $x = e^t, y = 2e^{2t} - 1$.

a. $y = 2x^2 - 1$ **for** $x > 0$ ✓

b. $y = 2x - 1$

c. $y = e^{2x} - 1$

d. $y = 4x^2 - 1$

$e^{2t} = (e^t)^2 = x^2$. Since $x = e^t > 0$, $y = 2x^2 - 1$ with domain $x > 0$.

27 [Foundational] • Target: Eliminating the Parameter (Trigonometric)

Eliminate the parameter from $x = 3\cos\theta, y = 3\sin\theta$.

a. $x^2 + y^2 = 9$ ✓

b. $x^2 + y^2 = 3$

c. $9x^2 + 9y^2 = 1$

d. $\frac{x^2}{9} - \frac{y^2}{9} = 1$

$\cos\theta = x/3, \sin\theta = y/3$. Using $\cos^2\theta + \sin^2\theta = 1$: $(x/3)^2 + (y/3)^2 = 1 \Rightarrow x^2 + y^2 = 9$.

28 [Foundational] • Target: Eliminating the Parameter (Trigonometric)

Identify the Cartesian equation for $x = 4\cos\theta, y = 2\sin\theta$.

a. $\frac{x^2}{16} + \frac{y^2}{4} = 1$ **(an ellipse)** ✓

b. $x^2 + y^2 = 20$

c. $\frac{x^2}{4} + \frac{y^2}{16} = 1$

d. $\frac{x^2}{16} - \frac{y^2}{4} = 1$

$(x/4)^2 + (y/2)^2 = \cos^2\theta + \sin^2\theta = 1 \Rightarrow \frac{x^2}{16} + \frac{y^2}{4} = 1$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Eliminating the Parameter (Trigonometric)**

What curve is traced by $x = 3\sec\theta, y = 2\tan\theta$?

a. **A hyperbola:** $\frac{x^2}{9} - \frac{y^2}{4} = 1$ ✓

b. An ellipse: $\frac{x^2}{9} + \frac{y^2}{4} = 1$

c. A parabola: $y = 2x^2$

d. A circle: $x^2 + y^2 = 13$

Using the identity $\sec^2\theta - \tan^2\theta = 1$: $(x/3)^2 - (y/2)^2 = 1 \Rightarrow \frac{x^2}{9} - \frac{y^2}{4} = 1$.

30 **[Intermediate]** • **Target: Projectile Motion Applications**

State the standard parametric equations for projectile motion launched with speed v_0 at angle θ from $(0, h_0)$ (with $g = 32$ ft/s^2).

a. $x(t) = (v_0\cos\theta)t, y(t) = -16t^2 + (v_0\sin\theta)t + h_0$ ✓

b. $x(t) = v_0t, y(t) = -32t^2 + h_0$

c. $x(t) = (v_0\sin\theta)t, y(t) = -16t^2 + h_0$

d. $x(t) = (v_0\cos\theta)t, y(t) = (v_0\sin\theta)t$

Horizontal velocity is constant ($v_0\cos\theta$), and vertical position is governed by gravity: $y(t) = -\frac{1}{2}gt^2 + v_{0y}t + h_0$.

31 **[Intermediate]** • **Target: Projectile Motion Applications**

A ball is kicked from ground level at 64 ft/s at an angle of 30°. At what time does it hit the ground?

a. 2 seconds ✓

b. 1 second

c. 4 seconds

d. 3 seconds

$v_{0y} = 64\sin(30^\circ) = 32$ ft/s. $y(t) = -16t^2 + 32t = -16t(t-2) = 0 \Rightarrow t = 2$ s.

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32 [Intermediate] • Target: Projectile Motion Applications

In the previous question, what is the horizontal range of the ball?

a. $64\sqrt{3}$ ft ✓

b. 128 ft

c. $32\sqrt{3}$ ft

d. 64 ft

$x(t) = (64\cos 30^\circ)t = (32\sqrt{3})(2) = 64\sqrt{3}$ ft.

33 [Intermediate] • Target: Projectile Motion Applications

Eliminate the parameter: $x = 2 + 5\cos t$, $y = 0 + 5\sin t$.

a. $(x-2)^2 + (y-0)^2 = 5$

b. $(x-2)^2 + (y-0)^2 = 25$ ✓

c. $(x-0)^2 + (y-2)^2 = 25$

d. $\frac{(x-2)^2}{25} + \frac{(y-0)^2}{25} = 1$

Since $\cos^2 t + \sin^2 t = 1$: $\left(\frac{x-2}{5}\right)^2 + \left(\frac{y-0}{5}\right)^2 = 1$, giving $(x-2)^2 + (y-0)^2 = 25$.

34 [Intermediate] • Target: Eliminating the parameter

Eliminate the parameter: $x = 0 + 4\cos t$, $y = -1 + 4\sin t$.

a. $\frac{(x-0)^2}{16} + \frac{(y+1)^2}{16} = 1$

b. $(x+1)^2 + (y-0)^2 = 16$

c. $(x-0)^2 + (y+1)^2 = 4$

d. $(x-0)^2 + (y+1)^2 = 16$ ✓

Since $\cos^2 t + \sin^2 t = 1$: $\left(\frac{x-0}{4}\right)^2 + \left(\frac{y+1}{4}\right)^2 = 1$, giving $(x-0)^2 + (y+1)^2 = 16$.

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35 [Intermediate] • Target: Projectile motion

Eliminate the parameter: $x = 0 + 3\cos t$, $y = -1 + 3\sin t$.

a. $(x-0)^2 + (y+1)^2 = 9$ ✓

b. $(x-0)^2 + (y+1)^2 = 3$

c. $\frac{(x-0)^2}{9} + \frac{(y+1)^2}{9} = 1$

d. $(x+1)^2 + (y-0)^2 = 9$

Since $\cos^2 t + \sin^2 t = 1$: $\left(\frac{x-0}{3}\right)^2 + \left(\frac{y+1}{3}\right)^2 = 1$, giving $(x-0)^2 + (y+1)^2 = 9$.

36 [Intermediate] • Target: Ellipses and other parametric curves

Eliminate the parameter: $x = 2 + 2\cos t$, $y = -1 + 2\sin t$.

a. $(x-2)^2 + (y+1)^2 = 4$ ✓

b. $(x-2)^2 + (y+1)^2 = 2$

c. $\frac{(x-2)^2}{4} + \frac{(y+1)^2}{4} = 1$

d. $(x+1)^2 + (y-2)^2 = 4$

Since $\cos^2 t + \sin^2 t = 1$: $\left(\frac{x-2}{2}\right)^2 + \left(\frac{y+1}{2}\right)^2 = 1$, giving $(x-2)^2 + (y+1)^2 = 4$.

37 [Intermediate] • Target: Finding the parametric derivative

Eliminate the parameter: $x = t$, $y = -1t + 5$.

a. $x = -1y + 5$

b. $y = -1x$

c. $y = -1x + 5$ ✓

d. $y = 1x + 5$

Substitute $t = x$ into the equation for y: $y = -1x + 5$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: More parametric practice • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Two vectors are given by $\vec{u} = 3\hat{i} + k\hat{j}$ and $\vec{v} = -\hat{i} + 4\hat{j}$. If $\vec{u}$ and $\vec{v}$ are orthogonal, find the value of the scalar parameter k.

a. 3/4

b. **4/3** ✓

c. -3/4

d. -4/3

Two vectors $\vec{u}$ and $\vec{v}$ are orthogonal if and only if their dot product is zero. The dot product of $\vec{u} = u_1\hat{i} + u_2\hat{j}$ and $\vec{v} = v_1\hat{i} + v_2\hat{j}$ is given by $\vec{u} \cdot \vec{v} = u_1v_1 + u_2v_2$. Given $\vec{u} = 3\hat{i} + k\hat{j}$ and $\vec{v} = -\hat{i} + 4\hat{j}$, their dot product is: $\vec{u} \cdot \vec{v} = (3)(-1) + (k)(4)$ $\vec{u} \cdot \vec{v} = -3 + 4k$ For the vectors to be orthogonal, their dot product must be zero: $-3 + 4k = 0$ $4k = 3$.

39 [Challenging] • Target: Graphing a parametric curve after elimination • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Solve $\cos(2x) + \sin x = 0$ on $[0, 2\pi)$.

a. $x = \frac{7\pi}{6}, \frac{11\pi}{6}$ only

b. $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$ ✓

c. $x = \frac{\pi}{2}$ only

d. $x = 0, \pi$

Using $\cos(2x) = 1 - 2\sin^2 x$: $1 - 2\sin^2 x + \sin x = 0 \Rightarrow 2\sin^2 x - \sin x - 1 = 0$. Factoring: $(2\sin x + 1)(\sin x - 1) = 0$, so $\sin x = 1$ or $\sin x = -\frac{1}{2}$. This gives $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$.

40 [Challenging] • Target: Restricting the parameter domain • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Solve $\cos(2x) + \sin x = 0$ on $[0, 2\pi)$. (attempt 3)

a. $x = \frac{\pi}{2}$ only

b. $x = \frac{7\pi}{6}, \frac{11\pi}{6}$ only

c. $x = 0, \pi$

d. $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$ ✓

Using $\cos(2x) = 1 - 2\sin^2 x$: $1 - 2\sin^2 x + \sin x = 0 \Rightarrow 2\sin^2 x - \sin x - 1 = 0$. Factoring: $(2\sin x + 1)(\sin x - 1) = 0$, so $\sin x = 1$ or $\sin x = -\frac{1}{2}$. This gives $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$.

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41 [Challenging] • Target: Parametrizing a line segment • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Solve $\cos(2x) + \sin x = 0$ on $[0, 2\pi)$. (attempt 6)

a. $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$ ✓

b. $x = \frac{\pi}{2}$ only

c. $x = 0, \pi$

d. $x = \frac{7\pi}{6}, \frac{11\pi}{6}$ only

Using $\cos(2x) = 1 - 2\sin^2 x$: $1 - 2\sin^2 x + \sin x = 0 \Rightarrow 2\sin^2 x - \sin x - 1 = 0$. Factoring: $(2\sin x + 1)(\sin x - 1) = 0$, so $\sin x = 1$ or $\sin x = -\frac{1}{2}$. This gives $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$.

42 [Challenging] • Target: Parametric speed and motion review • Cross-Lesson: 3.03 Sum and Difference Formulas

Solve $\cos(2x) + \sin x = 0$ on $[0, 2\pi)$. (attempt 7)

a. $x = 0, \pi$

b. $x = \frac{7\pi}{6}, \frac{11\pi}{6}$ only

c. $x = \frac{\pi}{2}$ only

d. $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$ ✓

Using $\cos(2x) = 1 - 2\sin^2 x$: $1 - 2\sin^2 x + \sin x = 0 \Rightarrow 2\sin^2 x - \sin x - 1 = 0$. Factoring: $(2\sin x + 1)(\sin x - 1) = 0$, so $\sin x = 1$ or $\sin x = -\frac{1}{2}$. This gives $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$.

43 [Challenging] • Target: Plotting and Direction of Motion • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Find the projection of $\mathbf{u} = \langle 6, -2 \rangle$ onto $\mathbf{v} = \langle 2, -1 \rangle$.

a. $\left\langle \frac{33}{5}, -\frac{14}{5} \right\rangle$

b. $\left\langle -\frac{14}{5}, \frac{28}{5} \right\rangle$

c. $\frac{14}{5}$

d. $\left\langle \frac{28}{5}, -\frac{14}{5} \right\rangle$ ✓

$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2}\mathbf{v} = \frac{14}{5}\langle 2, -1 \rangle = \left\langle \frac{28}{5}, -\frac{14}{5} \right\rangle$.

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44 [Challenging] • Target: Eliminating the Parameter (Algebraic) • Cross-Lesson: 4.01 Circles and Parabolas

Find the projection of $\mathbf{u} = \langle 3, 0 \rangle$ onto $\mathbf{v} = \langle 1, -1 \rangle$.

a. $\left\langle \frac{5}{2}, -\frac{3}{2} \right\rangle$

b. $\frac{3}{2}$

c. $\left\langle \frac{3}{2}, -\frac{3}{2} \right\rangle$ ✓

d. $\left\langle -\frac{3}{2}, \frac{3}{2} \right\rangle$

$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2}\mathbf{v} = \frac{3}{2}\langle 1, -1 \rangle = \left\langle \frac{3}{2}, -\frac{3}{2} \right\rangle$.

45 [Challenging] • Target: Eliminating the Parameter (Trigonometric) • Cross-Lesson: 4.02 Ellipses

Find the projection of $\mathbf{u} = \langle 3, 2 \rangle$ onto $\mathbf{v} = \langle 2, 2 \rangle$.

a. $\left\langle \frac{3}{2}, \frac{5}{2} \right\rangle$

b. $\left\langle \frac{5}{2}, \frac{5}{2} \right\rangle$ ✓

c. $\left\langle \frac{7}{2}, \frac{5}{2} \right\rangle$

d. $\frac{5}{4}$

$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2}\mathbf{v} = \frac{10}{8}\langle 2, 2 \rangle = \left\langle \frac{5}{2}, \frac{5}{2} \right\rangle$.

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